Valen Li

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Education

Julia R. Masterman High School Philadelphia, PA
GPA: 4.53/4.0 *Sep 2018 – June 2026*
Robotics Team (Co-Founder & Lead Programmer) | Chess Team (Co-Captain) | AI Club (President) | Programming Club (President)

The Wharton School of the University of Pennsylvania Remote
GPA: UW- 4.0/4.0 *Sep 2024 – Mar 2025*
STAT 0001: Intro to Statistics and Data Science

Community College Of Philadelphia In-Person
Courses: MATH 163: Discrete Mathematics I — Math 263 Discrete Mathematics II *Sep 2025 – Jun 2025*

The Wharton School of the University of Pennsylvania Residential
3-week residential program at UPenn called: The Leadership in the Business World *Jul 2025 – Aug 2025*

Oberlin College Remote
Pioneer Research Program (Under Full-Scholarship and given Four College Credits) *Feb 2025 – May 2025*

The Coding School Remote
Courses: Artificial Intelligence *Dec 2022 – May 2023*

Experience

Comcast Philadelphia, PA
Software Engineering Intern Jun 2025 - Aug 2025
- Developed a GitHub Actions workflow to automate MuleSoft project generation, reducing manual errors and saving developer time
- Orchestrated a CI/CD pipeline integrating GitHub Actions with Concourse for automated deployment
- Engineered a Flask frontend as a user interface, allowing users to configure the pipeline with a single click

Wharton's Stevens Center for Innovation in Finance Philadelphia, PA
Full-Stack App Developer Jun 2024 – Present
- Enhanced key features of a student loan application to accurately project college costs, using PostgreSQL for data management and Spring Boot for API development
- Designed a responsive website for the Finiverse project, and created web features for a data visualization platform.

Root2Success Philadelphia, PA
Co-President Feb 2022 - Present
- Tutored over 50 hours and developed comprehensive curriculums to teach mathematics to over 250 students across more than 25 partner schools in the School District of Philadelphia.
- Led a team of over 80 high school volunteers and organized in-person competitions to promote engagement and foster a love for mathematics.

Carnegie Mellon University Remote
Cyber-Security Researcher Feb 2025 – Present
- Worked with professor Hanan Hibshi in the research concentration of: The Practical Challenge of Writing Secure Code
- Focussed on the applications of Self-Adapting Machine Learning Models (SEAL) in cyber security, specifically phishing models
- Created a SEAL model that outperformed traditional classification models and wrote a paper

University of Washington Remote
Machine Learning Researcher June 2024 – Aug 2024
- Trained ML models such as XGBoost, LightGBM, CatBoost, Random Forest, K-Means Clustering, and GNNs on an emerging alternative (SMBJ) to traditional PCR techniques used in genetic sequencing and identification.
- Based on the results, on COVID Variants my hyper-parameterized model consistently achieved 99.9%+ accuracy even for single mismatch molecules.

Datacurve (YC W24) Remote
Machine Learning Model Trainer Nov 2024 – Feb 2024

- Trained Machine Learning Models for large companies and startups in need of ML models
- Helped develop the dataset used by these companies, utilized in training
- Primarily focused on using applicational math, and problem-solving skills to "teach" and train these models

Extracurricular Activities

FIRST Robotics Philadelphia, PA
Lead Programmer | Mentor | Inspection Lead | Tournament Volunteer | FRC Co-Founder Sep 2022 – Present

- Lead Programmer of teams: 22284 and 22279 (state-qualifier) simultaneously
- Co-founded FRC team 10918 Bluesteel Dragons first year at Masterman since 2019
- Raised $20k in funding to support new FRC team
- Created an autonomous program utilizing Virtual Field Mapping, Computer Vision, and Pure Pursuit algorithms
- Volunteered at FIRST Lego League (FLL) tournaments and mentor my school's FLL robotics team weekly
- Winner of: RoboJawn 2024 | Motivate Award | Think Award

USChess Onsite
Masterman Chess Team Co-Captain | Tournament Director | Chess Camp Instructor Sep 2018 – Present

- Co-Captain for the Elementary/Middle/High School Masterman Chess team
- Held weekly coaching sessions for the Elementary and Middle School
- Certified tournament director | Volunteered to direct the Elementary/Middle School Chess League
- Earned: >$3,500 | 2x @ World Open (4th) | 3x @ Nationals (5th) | Liberty Bell (1st) | Chess Congress (4th)

We Connect The Dots Remote
Community Ambassador | AI Summer Camp Instructor Dec 2023 – July 2025

- Managed sponsor outreach, program planning, and email marketing.
- Created a Free AI Summer Camp teaching how to use the AI.
- Taught students over the summer how to use AI libraries like Pandas, NumPy, and Matplotlib.

Artificial Intelligence (AI) Club Philadelphia, PA
President Sep 2021 - Present

- Taught Python and key AI libraries, and developed and led an AI summer camp in partnership with a non-profit organization.
- Educated over 75 members on foundational AI concepts and challenged common misconceptions about artificial intelligence.
- Secured 2nd place ($2500) at the Ctrl+Shift competition by creating and presenting li-valen.github.io/LearnAI.

Projects

Micro Pantry Mobile App | *Computer Vision, React-native, Gen AI* April 2024 – Present

- Developed an app using Expo using React and React-native as the frontend
- Implemented Tensorflow, specifically ICR, receipt, and food detection models to import into virtual Pantry
- Implemented Generative AI models to give feedback for meal suggestions, and nutrition advice

LearnAI | *Computer Vision, Javascript, GSAP* Feb 2024 – May 2024

- Created an award-winning website redefining AI education with $2,500 in funding.
- Addressed negative stigmas by providing interactive and comprehensive AI learning experiences.
- Developed a complex platform leveraging advanced models like CocoSsd and MobileNet.
- Built from scratch utilizing GSAP and Lenis for animations and smooth-scrolling also mimicking features from Apple's website.

Company Case Competition Analysis | *Figma, Business Strategy* Jul 2025 – Aug 2025

- Developed a comprehensive business strategy and pitch for Merck, recommending a partnership to create a complementary product for the newly acquired drug, Ohtuvayre.
- Designed and built a prototype for a smart inhaler and tracking app using Figma, visualizing the proposed product and user experience.
- Presented the pitch in a 15-minute presentation for the Leadership in the Business World case competition.

Real-time Object Detection with Edge Derivatives | *Python, OpenCV, PyTorch, YOLOv8, LaTeX* May 2025

- Authored a mathematical paper on image derivatives (Sobel, Laplacian) and implemented OpenCV edge detectors with code examples.
- Built a real-time object detection and instance segmentation pipeline using YOLOv8 with FPS overlay and COCO class labels.
- Documented setup and usage; provided a reproducible environment and compiled LaTeX paper.

UWash Research- SMBJ Classifier | *Python, CatBoost, LightGBM, XGBoost, scikit-learn* Jun 2024 – Aug 2024

- Built an AI system to classify COVID-19 variants from Single Molecule Break Junction (SMBJ) conductance signatures; achieved up to 99.97% accuracy (CatBoost).
- Implemented and compared 8 ML approaches (CatBoost, LightGBM, Random Forest, XGBoost, CNN+XGBoost, etc.) with rigorous sample-size analysis.
- Developed a preprocessing pipeline: filtering, R^2-based QC, conductance conversion, and 1D/2D histogram features.

The Woodlands Cemetery Project | *Web Development, Research* Mar 2025 – May 2025

- Researched and unearthed the life of James Hepburn Campbell, a 19th-century individual, as part of a rigorous research project for the A.P. United States History course.
- Independently created a professional website in just 3 days to detail Campbell's life and document the team's research process.
- Contributed over 280 hours to the project, demonstrating strong commitment and research skills.

M3 Mathworks Math Modeling Paper | *Mathematical Modeling, Data Analysis* Mar 2025 – Mar 2025

- Collaborated with a team during a 14-hour challenge to develop mathematical models for predicting key metrics during heatwaves.
- Engineered models to accurately predict indoor temperatures, peak power consumption, and a heat risk index.
- Analyzed complex data sets to inform modeling decisions and provide data-driven insights.

CMU Research on Cybersecurity | *AI, Cybersecurity, Research* Jun 2025 – Aug 2025

- Contributed to a research project on cybersecurity, focusing on using Self-Adapting Language Models (SEAL) for improving email phishing filters.
- Developed a research proposal and methodology for creating an AI-powered filter that can autonomously learn from new phishing tactics.
- Authored a paper, "Harnessing Self-Adapting Language Models (SEAL) for Continual Self-Improvement of Email Phishing Filters," detailing the project's findings and proposed solutions.

Awards

Good Citizen Award Mar 2025

Awarded by The Union League of Philadelphia for being 2/40,000 nominated by a Philly nonprofit for demonstrating hard work, honesty, curiosity, and loyalty

1st Place - Innovation League 2024 Dec 2024

Awarded $5000 | Prototyped, developed, and pitched an innovative tech startup to a panel of judges.

2nd Place - Ctrl+Shift 2024 May 2024

Awarded $2500 | Developed an innovative website that redefines AI education.

3rd Place- Penn Mathmatics Contest April 2025

Awarded by The University of Pennsylvania for winning 3rd Place, scoring a 65/80 at the Penn Math Contest (PMC IV)

Invited to the American Invitational Mathematics Examination Nov 2024

Scored a 96 (Top 5-10%) on the American Mathematics Competition (12B)

Congressional App Challenge Finalist Nov 2024

Recognized by the House of Representatives and Madeleine Dean

Stokes Grant Jun 2025

Awarded $1000 by Germantown Friends School for our startup

3rd Place- Bear Innovation Challenge Jun 2025

Awarded $1000 for pitching a DIY STEM Education company

Questbridge Prep Scholar May 2024

Recognized by Questbridge for academic excellence, personal qualities, and financial need

Top 150 M3 Mathworks Math Modeling Challenge Mar 2025

Recognized by the Society for Industrial and Applied Mathematics

National First Generation Recognition Program Aug 2024

Recognized by Collegboard for outstanding academic achievement as a first-generation student

National Recognition Program Jun 2025

Recognized by Collegboard for outstanding academic achievement